

TARAZ, Kazakhstan

WMO: 383410

Lat: 42.850N

Lon: 71.300E

Elev: 652

StdP: 93.73

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.5	-17.0	-22.9	0.5	-20.5	-19.2	0.7	-16.5	15.7	5.9	12.2	5.4	1.5	190	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	36.1	17.7	34.8	17.5	33.2	17.1	19.6	31.0	18.9	30.6	18.2	29.6	3.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.4	22.1	15.1	11.6	22.2	14.2	10.9	22.4	58.7	30.9	56.4	30.6	54.3	29.8	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	8.8	6.8	DB	-23.3	39.3	5.1	1.6	-27.0	40.4	-30.0	41.4	-32.9	42.2	-36.7	43.4	(4)
(5)			WB	-23.3	21.1	5.0	1.2	-26.9	22.0	-29.8	22.7	-32.6	23.3	-36.3	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-2.8	-1.2	6.1	12.4	17.9	23.1	25.1	23.7	18.1	11.5	4.2	-1.2	(6)	
	DBStd	11.11	6.57	6.76	5.83	4.75	3.69	3.29	2.79	3.19	4.23	5.06	6.28	6.65	(7)	
	HDD10.0	1472	397	316	146	29	1	0	0	0	1	43	190	349	(8)	
	HDD18.3	3127	654	547	381	183	52	4	1	3	55	217	424	607	(9)	
	CDD10.0	2015	1	2	24	102	247	394	468	426	244	90	16	1	(10)	
	CDD18.3	629	0	0	1	6	40	148	211	171	48	5	0	0	(11)	
	CDH23.3	8018	0	0	9	96	510	1856	2696	2151	615	82	5	0	(12)	
	CDH26.7	3763	0	0	1	20	156	873	1398	1066	228	22	0	0	(13)	
(14) Wind	WSAvg	2.7	2.4	2.4	2.8	3.0	3.1	3.0	2.9	3.0	2.7	2.3	2.3	2.5	(14)	
(15) Precipitation	PrecAvg	334	27	30	38	49	40	26	10	5	9	28	37	33	(15)	
	PrecMax	597	139	106	111	121	119	148	51	45	59	72	114	84	(16)	
	PrecMin	172	2	5	9	0	4	0	0	0	0	0	0	1	(17)	
	PrecStd	94	23	19	22	27	28	30	12	9	12	19	24	23	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	17.3	24.2	29.2	32.9	37.2	39.0	38.0	34.2	29.8	23.3	15.5	(19)	
		MCWB	7.5	8.7	12.0	15.8	16.6	17.9	18.0	17.8	16.6	14.5	12.2	7.5	(20)	
	2%	DB	10.7	13.0	20.8	26.1	30.2	35.0	36.5	35.7	31.5	25.2	18.8	12.0	(21)	
		MCWB	5.4	6.7	11.3	14.9	16.4	17.4	18.0	17.4	15.6	13.1	10.2	6.3	(22)	
	5%	DB	7.6	9.9	17.9	23.3	28.2	33.3	34.9	34.1	29.2	22.7	15.7	9.2	(23)	
		MCWB	3.9	5.4	10.1	14.1	16.0	17.1	17.6	17.2	14.9	11.8	9.0	4.9	(24)	
	10%	DB	5.1	7.1	15.0	20.9	26.2	31.8	33.2	32.3	27.0	20.1	12.9	6.9	(25)	
		MCWB	2.4	4.0	9.3	13.1	15.5	16.8	17.4	16.8	14.2	10.9	7.8	3.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	9.6	13.6	17.9	19.4	20.4	20.6	20.4	17.7	14.8	13.2	8.3	(27)	
		MCDB	13.4	16.0	21.1	25.2	27.9	32.2	32.2	32.3	30.8	26.4	19.5	13.1	(28)	
	2%	WB	5.8	7.7	11.9	16.1	18.2	19.1	19.6	19.0	16.5	13.6	11.2	6.8	(29)	
		MCDB	9.9	12.2	18.4	23.1	26.9	30.4	31.2	32.0	29.1	23.3	17.5	11.1	(30)	
	5%	WB	4.1	5.7	10.7	14.9	17.2	18.4	19.0	18.2	15.5	12.6	9.6	5.2	(31)	
		MCDB	7.1	9.2	16.7	21.8	25.5	29.3	31.0	30.7	27.6	20.8	15.1	8.6	(32)	
	10%	WB	2.5	4.1	9.5	13.7	16.3	17.7	18.3	17.5	14.5	11.6	8.0	3.6	(33)	
		MCDB	4.9	7.1	14.6	19.8	24.1	28.3	30.2	29.7	25.9	18.7	12.0	6.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	10.3	11.4	13.2	14.2	14.5	15.2	14.3	12.0	9.6	8.9	(35)	
		MCDBR	12.3	12.7	14.9	15.7	16.0	16.5	16.5	16.7	16.4	15.8	13.9	12.7	(36)	
	5% WB	MCWBR	8.0	7.7	7.6	7.6	6.5	5.6	5.4	5.8	6.6	7.2	7.4	7.9	(37)	
		MCWBR	11.1	11.8	13.3	13.3	13.6	14.1	14.3	15.1	15.5	13.6	12.5	11.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.333	0.370	0.434	0.453	0.434	0.425	0.416	0.385	0.365	0.399	0.381	0.326	(40)	
		taud	2.154	2.082	1.990	2.005	2.128	2.189	2.221	2.298	2.306	2.107	2.091	2.167	(41)	
	Ebn at Noon		771	802	794	814	842	850	853	870	861	758	698	739	(42)	
		Edh at Noon	99	126	157	167	152	144	138	123	113	122	106	89	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.86	2.71	3.63	4.91	6.39	7.25	7.29	6.72	5.29	3.25	1.90	1.50	(44)	
	RadStd		0.19	0.31	0.30	0.57	0.45	0.37	0.39	0.30	0.27	0.39	0.26	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.14	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page